

Divya Mehul Rajparia

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Summary

I am a **U.S. Citizen** and a CS graduate from IIT Hyderabad (top ~0.5% nationally). My background spans both AI research and product engineering: from domain adaptation research for tumor segmentation at USC to building end-to-end AI products and agentic systems using modern LLM frameworks. I'm comfortable taking ideas from research to production-grade software and quickly adapting to new tools, stacks, and problem domains.

Education

Degree	University/Institute	Year	CGPA/Marks(%)
B.Tech in Computer Science and Engineering	IIT Hyderabad	2026	9.44
XII High School (Senior Secondary)	Pace Junior Science College	2022	91.50%
X (Secondary School)	Lokhandwala Foundation School	2020	98.17%

Experience

Machine Learning Research Intern - Lab for Machine Learning and Biomedicine <i>University of Southern California, Los Angeles, CA Submitted to MICCAI 2026 Conference</i>	<i>May 2025 – Present</i>
- Automated tumor segmentation for breast ultrasound images (BUSI, BUS-UCLM), reducing reliance on manual annotation. - Addressed domain shift across medical datasets using a deep learning-based CycleGAN style transfer for semantic alignment. - Built an object-oriented modular PyTorch pipeline for training, checkpointing, evaluation, and metric logging. - Improved segmentation performance by +11.21% DSC and +15.81% IoU.	

Projects

DES: A Smarter Way to Grow ML Models Over Time (Python, PyTorch) Preprint <i>Machine Learning and Vision Lab, IIT Hyderabad</i>	<i>Jan 2025 – Present</i>
- Proposed AFCIL (Assumption-Free Class-Incremental Learning) and introduced Dynamic Expert Saturation (DES), a two-stage framework addressing catastrophic forgetting and long-tailed class imbalance in class-incremental learning. - Implemented in PyTorch with a frozen CLIP backbone and MLP adapters for adapter-only training. - Implemented class-aware sampling and optimized data loaders for long-tailed datasets (CIFAR100LT / ImageNetLT). - Benchmarked DES against strong baselines, achieving up to +6.38% accuracy improvements on CIFAR100LT and ImageNetLT.	
ExtBrew: AI-Native Chrome Extension Scaffolder (Next.js, TypeScript, Claude API) Project Link	<i>May 2026 – Present</i>
- Building a web app that scaffolds Chrome Manifest V3 extensions from natural-language prompts using Claude Sonnet 4.6. - Implemented token streaming by parsing Anthropic Server-Sent Events on a Next.js Edge route via TransformStream. - Engineered a tool-use loop where Claude invokes <code>create_file/edit_file/delete_file</code> tools to incrementally build a multi-file project visible in Monaco. - Designed a bring-your-own-key flow with localStorage persistence and an SSR-safe hydration hook to prevent React mismatches.	
Exploring Compositionality in Vision Transformers (Python, PyTorch) Preprint	<i>Jan 2025 – May 2025</i>
- Designed a framework using wavelet transforms to decompose images into sub-bands and test their linear composition in ViTs. - Demonstrated that ViTs exhibit compositionality despite strong non-linearity, and remain robust under noise and compression. - Evaluated robustness of ViT compositional representations on ImageNet under additive noise and image compression.	
Multi-Source Web Search Research Agent (Python, LangGraph, LLMs) GitHub	<i>Jan 2026 – Mar 2026</i>
- Built a multi-source research agent in LangGraph running parallel searches across Google, Bing, and Reddit. - Designed a stateful workflow with Pydantic-structured outputs to rank URLs via a zero-shot LLM, with BrightData SERP integration for async scraping. - Developed a modular synthesis pipeline using GPT-4o to combine results from multiple sources.	

Awards and Honors

- Academic Excellence Award (2024–2025):** Awarded for achieving the highest CGPA in the academic year.
- IUSSTF–Viterbi Scholar (2025):** Selected as one of the top 15 research students from across the country.
- Dr. Homi Bhabha Young Scientist Competition (Silver Medalist):** Ranked in the top **0.45%** of participants.
- JEE Mains (2022):** Scored **99.90 percentile** among 1 million+ candidates nationwide.

Extracurriculars

Ideation Manager, E-Cell IIT Hyderabad: Secured speakers, led ideation, and organized events.	<i>May 2023 – May 2024</i>
Head, Vibes Music Club IIT Hyderabad: Managed finances, events, media, and inductions.	<i>May 2023 – May 2024</i>

Tech Stack and Skills

Coding: Python, C/C++, SQL, HTML/CSS, Java	ML Frameworks: PyTorch, Scikit-Learn, TensorFlow
Tools & Platforms: Git, Linux, Jupyter, LangGraph	Data Science: NumPy, Pandas, Matplotlib
• Deep Learning • LLMs • Agentic AI • PyTorch • Transformers • Multimodal Learning • Model Architectures • Ranking Systems	